

IVIVC App User Guide

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1. Introduction

Establishing an in vitro-in vivo correlation (IVIVC) can support the evaluation of medical devices that include absorbable polymers by relating in vivo changes in material properties over time to corresponding in vitro test results, such as degradation profiles. IVIVCs may help evaluate manufacturing or material changes, support predictions of in vivo performance when modifying an existing device design, and reduce reliance on animal studies during preclinical development. This tool is intended to facilitate IVIVC development. The user supplies paired in vitro and in vivo fitting data, and may optionally supply an in vitro prediction dataset. The app applies selected preprocessing steps, fits and compares multiple correlation approaches and model families, evaluates selected performance metrics and cross-validation results, generates fitting and prediction plots, and produces report-ready summaries that can be reviewed and incorporated into external documentation. The app provides information to support user review of models. It does not select, recommend, validate, or establish regulatory acceptability for any model. Users should document their rationale for selecting a model.

For general background on credibility assessment and reporting of computational modeling studies in medical device submissions, see FDA guidance on computational modeling credibility and reporting [1, 2].

For additional discussion of model performance, cross-validation results, uncertainty, residual diagnostics, and model selection considerations, see Appendix A..

2. What the app does

- Applies selected normalization, scaling, and interpolation options to prepare fitting and prediction data.
- Fits user-selected IVIVC approaches and model families to paired in vitro and in vivo data.
- Calculates selected goodness-of-fit, model-comparison, and cross-validation metrics.
- Generates fitting, comparison, and prediction plots with uncertainty displays where supported.
- Creates summaries for user-selected final approaches or models.

3. Inputs

The IVIVC app requires paired in vitro and in vivo degradation data, along with user-selected analysis settings that define how the data are processed, modeled, and summarized. At a high level, the inputs include:

- a spreadsheet containing in vitro and in vivo time-course degradation data for fitting correlations;
- optionally, a spreadsheet containing in vitro data from which to generate predictions using fitted correlations;
- selection of one or more IVIVC modeling approaches;
- selection of candidate models to evaluate, where applicable;
- optionally, preprocessing options, such as normalization, interpolation, and scaling;
- selected performance metrics to display in the results and reports;
- optionally, cross-validation settings for applicable approaches; and
- grid-search settings for model initialization, where the options can usually be left to their default values.

The sections below provide additional detail on input file structure, required data columns, preprocessing options, model selection, and optional analysis settings.

3.1 Input datasets

The app uses separate fitting and prediction datasets. The fitting dataset is required because it provides the paired in vitro and in vivo data used to establish the IVIVC relationship. The prediction dataset is optional and is used only when applying a fitted relationship to new in vitro observations in order to predict the corresponding in vivo behavior.

In this guide, **response value** means the measured y-axis value that changes with time, such as mass remaining, molecular weight, strength, or another degradation-related endpoint. In the input files, these are the in vitro and in vivo value columns.

Input	Description
Fitting dataset	Required four-column dataset used to establish the in vitro/in vivo relationship. Columns are interpreted as in vitro time, in vitro response value, in vivo time, and in vivo response value.
Prediction dataset	Optional two-column in vitro dataset used after fitting to generate in vivo predictions. Columns are interpreted as in vitro time and in vitro response value.

Acceptable formats are CSV and Excel (.xls, .xlsx). For Excel workbooks, select the sheet containing the relevant data. For csv files, the first columns are read directly in the order described above.

3.2 Analysis approaches and models

In this app, an analysis **approach** is the overarching method used to establish or evaluate an in vitro/in vivo correlation. A **model** is the mathematical form used within an approach when an equation is needed to describe the relationship.

3.2.1 Approaches

Approaches differ in how the in vitro and in vivo time courses are related to one another.

Approach	Description	Available models
Approach 1: Direct Correlation	Approach 1 directly relates matched in vitro and in vivo time points after preprocessing (i.e., in vivo time is modeled as a function of in vitro time). The fitted model is then used as a time-scaling function that maps an in vitro time to a predicted in vivo time. This approach can be useful when the two datasets have matching time points, or when interpolation can create matched pairs, and the main question is how in vitro time corresponds to in vivo time.	Double Exponential, Exponential, Linear, Linear (zero intercept), Power
Approach 2: Independent Model Fitting	Approach 2 fits the same model family (e.g., exponential decay) separately to the in vitro and in vivo datasets. With fitting data alone, the report summarizes the two fitted models and uncertainty for those fitted curves. When prediction data are supplied, predictions can be generated by comparing the fitted in vivo and in vitro responses or by comparing fitted characteristic time constants when those values are available. This approach can be useful when the two datasets were measured on different time schedules, when each dataset is expected to follow the same type of degradation curve, or when a specific model form is scientifically meaningful for both datasets.	Double Exponential, Exponential, Mass Loss (Gompertz), Mass Loss (One-Phase), Mass Loss (Surface Erosion), Mass Loss (Two-Phase), Power, Stretched Exponential

Approach	Description	Available models
Approach 3: Direct Mapping	Approach 3 uses direct interpolation-based mappings between datasets rather than fitting a parametric model. It can describe response-value ratios over time and time ratios across response-value ranges. When prediction data are supplied, those mappings can be used to map in vitro response values or in vitro times to the corresponding in vivo response values or times. This approach can be useful for exploratory comparisons, for datasets where a simple equation does not describe the observed pattern well, or when the user wants a direct data-based mapping rather than a fitted model.	No parametric model selection

3.2.2 Models

Candidate models are the mathematical equations available for fitting within the applicable approaches. Different model forms represent different assumptions about the shape of the degradation relationship, such as linear behavior, power-law behavior, or sigmoidal transition behavior.

Model	Used in	Functional form	Notes
Double Exponential	Approach 1, Approach 2	$y = ae^{bx} + ce^{dx}$	Sum of two exponential terms. It can represent profiles with two apparent rates, but the added flexibility may require more data to estimate stably.
Exponential	Approach 1, Approach 2	$y = ae^{bx}$	Two-parameter exponential relationship. It can describe monotonic change that accelerates or decelerates exponentially over the fitted range.
Linear	Approach 1	$y = a + bx$	Straight-line empirical relationship with an intercept and slope. It is appropriate when the in vitro/in vivo relationship is approximately linear over the fitted range.
Linear (zero intercept)	Approach 1	$y = ax$	Straight-line empirical relationship constrained to pass through zero. It may be useful when a zero in vitro value should correspond to a zero in vivo value under the selected preprocessing and response scale.

Model	Used in	Functional form	Notes
Mass Loss (Gompertz)	Approach 2	$y = y_0 - A \exp\{-\exp[(er_{max}/A)(t_{lag} - x) + 1]\}$	Mass-remaining model based on the modified Gompertz curve. It is intended for asymmetric sigmoidal mass-loss profiles with an apparent lag or plateau, a maximum loss-rate region, and a terminal loss extent.
Mass Loss (One-Phase)	Approach 2	$y = y_0 - A/(1 + e^{-(x-t_{50})/s})$	Mass-remaining model with a delayed sigmoidal loss phase. It is intended for mass-loss profiles that begin constant and show a later drop.
Mass Loss (Surface Erosion)	Approach 2	$y = y_0 \max(1 - kx, 0)^n$	Mass-remaining model for surface-controlled erosion. The fitted exponent n allows the profile shape to adapt to slab-like, cylindrical, spherical, or empirical geometry-like erosion behavior.
Mass Loss (Two-Phase)	Approach 2	$y = y_0 - B(1 - e^{-k_b x}) - D/(1 + e^{-(x-t_{50})/s})$	Mass-remaining model with an optional early burst-loss phase followed by a delayed sigmoidal loss phase. It is intended for mass-loss profiles that may show an initial drop, an intermediate plateau, and a later drop.
Power	Approach 1, Approach 2	$y = ax^b$	Power-law empirical relationship. It can describe nonlinear monotonic relationships where the response changes as a fitted power of time or the mapped predictor.
Stretched Exponential	Approach 2	$y = ae^{bx^c}$	Exponential relationship with a fitted stretching exponent. It can describe non-single-rate behavior, but the additional shape parameter can increase sensitivity when data are sparse.

Mass loss model notes

The surface erosion mass loss form is based on geometry-controlled surface erosion concepts from the erodible slab, cylinder, and sphere model [3]. The one-phase mass loss model uses a logistic/sigmoidal transition to represent a delayed loss phase [4]. The two-phase mass loss model combines an early burst-like loss term with a delayed sigmoidal loss phase. Burst and delayed/multiphase behavior in degradable polymer systems is discussed in the polymer erosion and PLGA release literature [5, 6, 7]. The Gompertz mass loss model uses a modified Gompertz-style sigmoidal form with lag and maximum-rate parameters [8].

3.3 Preprocessing options

Preprocessing prepares the uploaded data for comparison, fitting, and prediction. Normalization expresses response values relative to a reference value so datasets can be compared on a common relative scale. Scaling changes the numerical scale of time or response values without changing the underlying pattern. Interpolation estimates response values at time points that were not directly observed, which may be useful when an approach requires matching in vitro and in vivo time points (i.e., Approach 1, Approach 3).

3.4 Normalization

Option	Internal key	Description
Min-max normalization	min_max	Converts response values to a 0-1 scale using the observed minimum and maximum within each dataset.
Initial-value normalization	initial_value	Expresses each response value relative to the first response value in the corresponding dataset.

3.5 Scaling

Option	Internal key	Description
Log10 time scaling	log_x	Uses a base-10 log scale for time values before fitting or interpolation.
Log10 response scaling	log_y	Uses a base-10 log scale for response values before fitting or interpolation.

3.6 Interpolation

Option	Internal key	Description
No interpolation	none	Uses only the original measured points. Response values between measurements are not estimated, so unmatched positions are marked as missing.
Shared-time interpolation	default	Uses linear interpolation to estimate response values for both datasets at a shared set of experimental time points. Estimates outside the measured time range are marked as unavailable.
Densified interpolation	alternative	Adds extra evenly spaced points between adjacent observations before interpolation. This can smooth displays and make fitting more stable, but the added response values are estimated rather than directly measured and can affect metric values.

3.7 Performance metrics

Performance metrics are summary measures that help describe how closely a fitted model matches the observed data (e.g., R^2), how large the prediction errors are (e.g., RMSE), and, for some metrics, how well a model balances fit quality with model complexity (e.g., information criteria). The following performance metrics are included in the app.

Metric	Internal key	Preferred direction	Description
R^2	r_squared	Higher is better	Coefficient of determination. Higher values indicate that the fitted curve follows more of the observed response pattern.

Metric	Internal key	Preferred direction	Description
Adjusted R^2	adjusted_r_squared	Higher is better	R^2 adjusted for the number of fitted parameters. It can decrease when extra parameters do not improve the fit enough to justify the added complexity.
MSE	mse	Lower is better	Average squared difference between observed and fitted values. Lower values indicate smaller model errors, but the squared units can be harder to interpret directly.
RMSE	rmse	Lower is better	Typical model error in the same response units as the fitted data. Lower values indicate predictions closer to the observations.
MAE	mae	Lower is better	Average absolute model error in the same response units as the fitted data. Lower values indicate predictions closer to the observations.
NRMSE (mean)	mnrmse	Lower is better	RMSE divided by the response mean. Lower values indicate smaller error relative to the typical response magnitude.
NRMSE (std. dev.)	nrmse	Lower is better	RMSE divided by the response standard deviation. Lower values indicate smaller error relative to the amount of variation in the response.
AIC	aic	Lower is better	Akaike Information Criterion. AIC compares fitted models by considering both fit quality and the number of fitted parameters. Among comparable fitted models, a lower AIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute AIC values, including negative values, should not be interpreted as standalone quality grades.
AIC _c	aicc	Lower is better	Corrected Akaike Information Criterion. AIC _c is AIC with an added small-sample correction and is often preferred when the number of fitted observations is limited relative to the number of fitted parameters. See the user guide for additional information. Among comparable fitted models, a lower AIC _c value generally indicates better empirical support for a model that balances fit quality and complexity; AIC _c is undefined when there are too few observations relative to the number of parameters.

Metric	Internal key	Preferred direction	Description
BIC	bic	Lower is better	Bayesian Information Criterion. BIC compares fitted models by considering both fit quality and the number of fitted parameters, usually applying a stronger penalty for extra parameters than AIC. Among comparable fitted models, a lower BIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute BIC values, including negative values, should not be interpreted as standalone quality grades.

For additional background on common error metrics, scale-dependent accuracy measures, and adjusted R^2 , see references [9, 10, 11].

For AIC_c , a rule of thumb is to prefer AIC_c over AIC when the ratio of fitted observations to estimated parameters is small, such as $n/k < 40$, where n is the number of fitted observations and k is the number of model parameters [12], which we have found to be the case for typical IVIVC datasets.

By default, the app selects R^2 , RMSE, and AIC_c . This default set has been useful in internal testing with real datasets, but it is not necessarily the best set for every dataset or intended use. Users may select additional metrics when they provide useful context for a specific analysis. The default set gives three complementary views of model performance. R^2 is widely known and describes how well the fitted curve follows the observed response pattern. RMSE reports typical error in the original response units, making it easier to judge whether the error is practically large or small. AIC_c supports comparison among fitted candidate models by balancing fit quality with a penalty for unnecessary model complexity, with a correction that is useful for smaller datasets.

3.8 Cross-validation options

Cross-validation (CV) checks whether model performance changes substantially when different observations are omitted from fitting. The app does this by repeatedly setting aside part of the dataset, fitting the model to the remaining observations, and comparing model predictions with the observations that were set aside.

In this app, CV should be interpreted as a **prediction-error diagnostic** and sensitivity check, not as a confidence interval or parameter uncertainty estimate. If the error from the model fitted to the full dataset is similar to the CV error, that provides reassurance that the fitted relationship is not overly dependent on a particular subset of observations. If the CV error is much worse, the model may be sensitive to the available data, the selected splitting scheme, or the number and placement of time points.

For time-course data, the splitting scheme matters because nearby observations may contain related information, and early, middle, and late time points may represent different parts of the degradation process. Setting aside one time point mainly asks whether the model can recover an individual omitted observation. Setting aside a contiguous block asks whether the model can bridge a missing region of the profile. Setting aside a random subset is a more general check of whether the fitted curve captures the overall response pattern, but it is less time-aware.

There is no universally best CV scheme for every dataset. The most useful choice depends on the question being asked and on whether enough observations remain for stable fitting. In this app, the goal is usually to assess whether a mathematical curve captures the overall degradation or correlation pattern, rather than to forecast future time points from earlier time points. When results change substantially across CV schemes, review the fitted plots, residual diagnostics, parameter uncertainty, model complexity, and scientific plausibility rather than treating CV as a pass/fail result.

For background on cross-validation procedures and their use in model assessment and selection, including issues that arise with time series, see references [13, 14, 15, 16].

3.8.1 Choosing a cross-validation scheme

The table below summarizes the schemes available in the app. The “CV error describes” column explains what kind of prediction sensitivity each scheme is most directly measuring.

Scheme	CV error describes	Useful when	Main caution
Leave-one-timepoint-out	How well the model predicts a single omitted time point at a time; how sensitive the fit is to setting aside one observation.	A good default starting point for small datasets, because each repeated fit still uses nearly all of the available data.	This can be a gentle test: if the time points on either side of the omitted one are close together, they may largely determine the prediction, making the model appear more robust than it is.
Leave-contiguous-block-out	How well the model predicts an entire missing time window; sensitivity to a gap in the observed profile.	The primary time-aware sensitivity check, especially when adjacent observations are likely to be related or when a missing time region is a practical concern.	Larger blocks make the test harder but may leave too few points for the model to fit stably. Start with small blocks and increase cautiously.
K-fold	How predictions change when larger ordered portions of the dataset are set aside; sensitivity to removing whole folds of time-ordered data.	A familiar fold-based diagnostic when there are enough observations to form meaningful folds. Smaller values of k create larger test regions and a harder test.	The training data for a given test fold may include time points both before and after it. Results can be sensitive to where the fold boundaries happen to fall.
Shuffle split	How robust overall curve predictions are to repeated random selections of training and test points.	A general robustness check of overall curve shape when time ordering is not the primary concern.	Because it ignores time order, this scheme can make a model appear more reliable than it is when nearby time points carry similar information. Leave-one-timepoint-out or contiguous-block CV is usually more informative for time-course data.
None	No cross-validation is performed; only the model fitted to the full dataset is summarized.	Appropriate when there are too few observations to split meaningfully, repeated fits are numerically unstable, or the goal is simply to examine the fitted model itself.	Removes one diagnostic tool. When CV is skipped, rely more carefully on fitted plots, residuals, final-fit metrics, parameter diagnostics, and scientific plausibility.

3.8.2 Cross-validation parameters

Parameter	Internal key	Description
Cross-validation scheme	cv_scheme	Selects the splitting method used to estimate cross-validation performance for fitted parametric models. Leave-one-timepoint-out is the default.
CV splits/folds/block size	cv_n_splits	For Shuffle split, the number of repeated random train/test splits. For K-fold, the number of folds. For Leave-contiguous-block-out, the number of adjacent time points set aside per split. The setup page does not enforce a maximum block size because the uploaded data are not read until Analyze is selected; larger blocks may leave too few observations for stable fitting. This setting is not used for Leave-one-timepoint-out or None.
Shuffle split test size	cv_test_size	The fraction of observations used as test data in each Shuffle split repeat. Smaller values leave more observations for fitting; larger values make the test more demanding but leave fewer observations for training.
CV random seed	cv_random_state	Seed used to make randomized cross-validation splits reproducible. This applies to Shuffle split.

3.9 Grid-search initialization

When the app fits a mathematical curve, it needs starting values for the model parameters. Poor starting values can lead to an unstable or unreasonable fit, especially for nonlinear models. Grid-search initialization reduces this risk by trying many candidate starting-value sets before nonlinear least-squares fitting.

The app evaluates those candidate starts, chooses the most promising one, and then runs the final curve-fitting step from that starting point. This improves robustness, but it does not guarantee that the fit is the global best solution.

The default grid-search settings can usually be left unchanged. If fitting fails or appears unstable, consider increasing the number of random starts, adjusting the parameter bounds, changing the random seed, or reviewing whether the selected model is appropriate for the data.

Parameter	Internal key	Description
Grid-search random starts	grid_search_num_points	Number of candidate starting parameter sets evaluated before nonlinear least-squares fitting. Higher values may improve robustness but increase runtime.
Grid-search CPU cores	grid_search_num_cores	Number of CPU cores used for the starting-point search. Only available via command-line interface.
Grid-search parameter minimum	grid_search_param_min	Lower bound for randomly sampled candidate starting parameter values.
Grid-search parameter maximum	grid_search_param_max	Upper bound for randomly sampled candidate starting parameter values.
Grid-search random seed	grid_search_random_state	Seed used to make randomized starting-point sampling reproducible. Changing the seed changes the sampled candidates while keeping the same bounds and number of starts.

4. Outputs

The IVIVC app generates interactive results and downloadable reports that summarize fitted model behavior, model-comparison metrics, diagnostic information, and user-selected analysis settings. Output availability depends on the selected approaches, whether fitting succeeds, and whether a prediction dataset was supplied.

Output	Description
Preprocessing outputs	Processed data tables and plots showing the data used for fitting and prediction.
Model fitting outputs	Parameter estimates reported as estimate $\pm$ 95% confidence interval where available, fitted plots with error bars, residual plots, and selected performance metrics.
Model comparison outputs	Tables comparing selected approaches, models, metrics, relative model evidence, and cross-validation summaries.
Prediction outputs	Prediction plots and downloadable tables when a prediction dataset is supplied and the selected method can be applied.
Selected model & summary of outputs	User-selected report summaries and optional report export for incorporation into external documentation.

4.1 Relative model evidence

Relative model evidence tables compare candidate parametric models using information criteria (i.e., AIC, AIC_c, and/or BIC), showing which models have more support from the observed data after balancing fit quality against model complexity. These comparisons are only meaningful among models fitted to the same data, with the same preprocessing settings, and within the same IVIVC approach.

These tables are shown only when one or more information criteria are selected as metrics. If more than one criterion is selected, review each separately—AIC, AIC_c, and BIC can occasionally rank the same models differently. See the appendix for interpretation guidance, rules of thumb, and limitations.

The relative model evidence tables contain the following quantities.

Table output	What it means
n	Number of observations used to fit the model.
k	Number of fitted model parameters. Models with more parameters can fit more flexible shapes, so information criteria include a complexity penalty.
AIC, AIC _c , or BIC	Information-criterion value for the fitted model. Lower values indicate stronger relative support within the candidate set.
Delta	Difference between a model's criterion value and the lowest value in the candidate set. Values near 0 indicate similar support to the leading model; larger values indicate less support.
Weight	Normalized relative-support score across the candidate set. Weights are comparative summaries, not absolute probabilities that a model is correct.
Evidence ratio	Ratio of the highest weight in the candidate set to the model's weight. Larger values indicate less relative support compared with the leading model.
Support badge	Rule-of-thumb label based on the delta value. Intended to aid user review, not to recommend a model.

4.2 Uncertainty and error bars

These displays show how much fitted curves or predictions may vary based on the fitted model, residual error, and preprocessing choices. They are intended to support review of uncertainty, not to define acceptance limits.

The following descriptions summarize how fitted-parameter uncertainty, fitting-plot bands, and prediction error bars are calculated or interpreted for each approach.

4.2.1 Approach 1: Direct Correlation

Fitting uncertainty

Parameter estimates: fitted parameters are reported as estimate $\pm$ 95% confidence interval. The plus-minus value is calculated from the nonlinear least-squares parameter covariance matrix as $t(0.975, \text{residual degrees of freedom})$

multiplied by the parameter standard error. Correlations among fitted parameters are retained when uncertainty-aware parameter values are created.

Fitting plot bands: shaded 95% prediction bands are generated by Monte Carlo sampling of the fitted parameter covariance matrix. The sampled model variance is combined with the residual variance from the fitted data, then converted to a two-sided interval using a t multiplier when residual degrees of freedom are available.

These bands describe approximate prediction uncertainty around the fitted relationship, not regulatory acceptance limits. They depend on the fitted model form, local covariance approximation, residual behavior, and preprocessing choices.

Prediction uncertainty

Prediction plot error bars: when a prediction dataset is supplied, predicted in vivo times are plotted with horizontal error bars. The bars are calculated from the same 95% prediction-band calculation used for the fitted correlation model; the upper and lower bar lengths are the distances from the band mean to the corresponding upper and lower interval bounds at each input prediction time.

The error bars therefore include propagated fitted-parameter uncertainty and residual/model error from the Approach 1 correlation fit.

4.2.2 Approach 2: Independent Model Fitting

Fitting uncertainty

Parameter estimates: each in vitro and in vivo model parameter is reported as estimate $\pm$ 95% confidence interval. The plus-minus value is calculated from the corresponding nonlinear least-squares covariance matrix as $t(0.975, \text{residual degrees of freedom})$ multiplied by the parameter standard error, with fitted-parameter correlations retained in uncertainty-aware values.

Time constants: tau values, when available, are calculated from the fitted model and parameter covariance. Reported tau values are shown as tau $\pm$ 95% confidence interval. The plus-minus value is calculated as $t(0.975, \text{residual degrees of freedom})$ multiplied by the propagated standard uncertainty.

Fitting plot bands: the in vitro and in vivo shaded 95% prediction bands are generated separately. For each fitted model, the app Monte Carlo samples the fitted parameter covariance matrix, combines sampled model variance with residual variance, and converts the resulting standard uncertainty to a two-sided interval using a t multiplier when residual degrees of freedom are available.

Prediction uncertainty

Response-value-ratio prediction error bars: uncertainty is propagated from both fitted models. The app samples each model's parameter covariance matrix, calculates the in vivo/in vitro response-value ratio for each sample, combines variance in that ratio with residual-based ratio uncertainty, and converts that uncertainty to a two-sided 95% interval. The plotted vertical error bars are scaled from the resulting ratio interval at each prediction time.

Time-constant-ratio prediction error bars: when both tau values are valid, uncertainty in each tau is propagated through the tau ratio using uncertainty-aware arithmetic. The standard uncertainty of the scaled prediction time is multiplied by a t value, where available, and plotted as a symmetric horizontal error bar.

Response-value ratio uncertainty can become unstable when the denominator model prediction or denominator time constant is near zero. Predictions in those regions should be interpreted cautiously.

4.2.3 Approach 3: Direct Mapping

Fitting uncertainty

Parameter uncertainty: Approach 3 does not fit a parametric model, so there is no fitted-parameter covariance matrix, no parameter confidence intervals, and no model-based 95% prediction band.

Fitting plots: the plots show interpolation-based ratios/mappings calculated from the preprocessed datasets. They should be interpreted as deterministic mappings from the supplied data and preprocessing choices, not as statistical uncertainty bands.

Prediction uncertainty

Prediction plot error bars: Approach 3 does not generate formal propagated error bars for prediction outputs because no parametric covariance model is fitted.

Uncertainty is therefore limited to the quality of the input data, preprocessing, interpolation, and the assumption that the observed mapping remains valid for the prediction dataset. Extrapolated points outside the interpolation range are excluded or returned as unavailable.

5. Command-line interface

The repository includes `cli.py` for batch-oriented fitting from the command line. Use `python cli.py --help` to see available arguments for input files, approaches, models, metrics, preprocessing options, cross-validation settings, and grid-search settings. The internal keys specified in this user guide are used as settings with `cli.py`.

6. Extending the app

The IVIVC app is designed so that common extensions can be added with minimal changes to the main application code. In many cases, new models, performance metrics, and preprocessing methods can be added by following the structure of the existing files in the relevant subdirectories.

Appendix

A. How to Interpret Model Performance

No single metric determines whether an IVIVC model is meaningful. A complete evaluation should consider goodness of fit, prediction error, model complexity, cross-validation behavior, residual patterns, parameter uncertainty, and whether the fitted relationship is scientifically plausible for the polymer degradation mechanism and intended use. Performance metrics are descriptive and comparative tools, not acceptance criteria. Small datasets can produce unstable estimates, and similar metric values do not necessarily indicate equivalent scientific validity. For analyses intended to support regulatory submissions, the depth of supporting evidence should be proportionate to the model's intended use and credibility requirements [1, 2].

The app reports metrics and comparison tables to support user review. It does not select, recommend, validate, or establish regulatory acceptability for any model. Users should document preprocessing assumptions, fitting settings, model selection rationale, limitations, and any planned use of the model outside the fitted data range.

A.1 Confirm that models are comparable before interpreting metrics

Performance metrics are most meaningful when the models being compared were evaluated under identical conditions. Only compare models within the same uploaded dataset, preprocessing configuration, response scale, selected IVIVC approach, and model set.

In particular, do not compare AIC, AIC_c, or BIC values across different datasets, transformed responses, preprocessing settings, or substantially different fitting contexts. These information criteria measure the *relative* support among models evaluated under the same conditions—they are not absolute quality scores and lose their meaning when the underlying data or setup differs. Approach 3 should also be interpreted differently from Approaches 1 and 2, because it uses direct interpolation-based mappings rather than fitting a parametric model.

A.2 Metric ranges, direction, and interpretation

Metric	Range / direction	Interpretation guidance
R^2	Best possible value is 1. Values near 0 indicate little improvement over predicting the mean response. Can be negative if the model performs worse than the mean-response baseline. Higher is better.	Proportion of response variance explained by the model [17]. High R^2 does not imply that the model is mechanistic, causal, predictive outside the fitted data range, or appropriate for the intended use.
Adjusted R^2	Higher is better. Can be lower than R^2 and may be negative.	Penalizes additional fitted parameters, making it more informative than R^2 when comparing models of different complexity [11].
MAE	Lower is better. Units match the response units.	Mean absolute error on the response scale [17]. Interpret relative to assay variability, measurement uncertainty, and the intended use.
MSE	Lower is better. Units are squared response units.	Mean squared prediction error [9, 17]. Useful computationally, but less intuitive than RMSE because the units are squared.
RMSE	Lower is better. Units match the response units.	Root mean squared error, the typical prediction error on the response scale [9, 17]. Interpret relative to assay variability, measurement uncertainty, and the intended use. Penalizes larger errors more heavily than MAE.
NRMSE	Lower is better. Normalized to allow more meaningful comparison across response scales.	RMSE prediction error expressed relative to a normalization factor such as the response mean or standard deviation. Interpretation depends on the normalization method used.

Metric	Range / direction	Interpretation guidance
AIC	Lower is better, but only relative to other comparable models. No fixed bounds; negative values are allowed.	Balances goodness of fit against model complexity [18, 12, 19]. A lower value indicates a model that explains the data well without unnecessary parameters. Absolute values are not standalone quality grades.
AIC _c	Lower is better, but only relative to other comparable models. Undefined when there are too few observations relative to the number of fitted parameters.	A small-sample correction to AIC [20, 12, 19]. Generally preferred when the observation-to-parameter ratio is low; $n/k < 40$ is a commonly cited rule of thumb [12].
BIC	Lower is better, but only relative to other comparable models. No fixed bounds; negative values are allowed.	Balances goodness of fit against model complexity, typically applying a stronger penalty for additional parameters than AIC [21, 12, 19].
Cross-validation metrics	Direction follows the underlying metric: higher is better for R^2 -type metrics, lower is better for error metrics.	Estimate prediction performance under the selected splitting scheme [13, 14, 15, 16]. Can help identify overfitting, instability, or sensitivity to individual observations or data splits.
Fit/CV ratio	Values near 1 indicate similar fitted and cross-validation performance.	A ratio near 1 suggests the model generalizes consistently. Substantial deviations may indicate overfitting, instability, or sensitivity to the splitting scheme, though no universal acceptance threshold exists.

A.3 Interpreting goodness of fit and prediction error

Goodness-of-fit metrics— R^2 , adjusted R^2 , MSE, RMSE, and NRMSE—describe how well the model fits the data used to estimate its parameters. These metrics should be interpreted in practical context, not just by their numerical value, and there is no universal performance cutoff for IVIVC adequacy.

Strong goodness-of-fit is encouraging, but can be misleading, particularly for small datasets or models with many parameters. A model with excellent fit metrics may still be unsuitable if its parameters are unstable, cross-validation performance is poor, residuals show systematic patterns, or the fitted curve shape is not scientifically plausible for the degradation process being modeled.

RMSE is often the most intuitive error metric because it is expressed in the same units as the response. NRMSE can be useful for comparing error across different response scales, but the choice of denominator changes what the metric actually measures [10]:

- **NRMSE normalized by the response mean** answers: “How large is the typical prediction error relative to the average response value?” This normalization is useful for comparing model error across endpoints with different scales or units. For example, comparing how well a model fits mass loss data versus molecular weight data on a common scale.
- **NRMSE normalized by the response standard deviation** answers: “How large is the typical prediction error relative to the natural spread of the observed responses?” This normalization is useful for gauging how much of the variability in the data the model fails to capture. A value approaching or exceeding 1 indicates that the typical prediction error is as large as the spread of the data itself, suggesting the model explains little of the observed variability, analogous to an R^2 near zero.

Take care when comparing NRMSE values across datasets or response scales if the normalization method differs. More broadly, whether a given error level is acceptable depends on the context: a numerically small error may still matter if the relevant degradation endpoint has narrow acceptable variation, while a larger error may be tolerable during exploratory model development when high predictive precision is not required.

A.4 Interpreting relative model evidence

Relative model evidence tables compare candidate models using information criteria (i.e., AIC, AIC_c, and BIC), which balance goodness of fit against model complexity. In this context, *empirical support* means that the observed data provide more support for one model than another within the displayed candidate set, based on the selected information criterion. It does not prove that a model is correct, mechanistic, predictive outside the observed range, or suitable for a regulatory submission.

Rules of thumb for delta values

Delta is the difference between a model's criterion value and the lowest value in the candidate set. Common interpretive conventions are:

- **AIC / AIC_c**: delta 0–2 suggests similar or substantial support; 4–7 suggests considerably less support; greater than 10 suggests essentially no support relative to the leading model [12].
- **BIC**: delta 0–2 suggests similar support; 2–6 suggests less support; 6–10 suggests substantially less support; greater than 10 suggests essentially no support relative to the leading model [22].

These are interpretive conventions, not regulatory acceptance criteria.

Evidence weights

For a given criterion, the evidence weight for model i is:

$$w_i = \frac{\exp\left(-\frac{1}{2}\Delta_i\right)}{\sum_{j=1}^J \exp\left(-\frac{1}{2}\Delta_j\right)} \quad (1)$$

where Δ_i is the delta value for model i and J is the number of candidate models. Larger weights indicate stronger relative support within the candidate set [12, 19].

When multiple models have similar support

If several models have similar delta values or weights, relative support alone is not sufficient to choose among them. Also consider model simplicity, parameter plausibility, residual behavior, cross-validation performance, and scientific interpretability.

A.5 Interpreting cross-validation performance

Cross-validation estimates how well the model predicts observations that were not used to fit it, under the chosen splitting pattern. In this app, it is most useful as a **sensitivity diagnostic**: when the cross-validation error is similar to the error from fitting the full dataset, this suggests the fitted relationship is reasonably stable regardless of which observations are set aside. When the two differ substantially, the model may be more sensitive to which specific data points were included in fitting, or to the particular way the data were split. Cross-validation does not measure uncertainty in the model parameters. The practical question to keep in mind is not just “how good is the score?” but also “what kind of prediction task did the cross-validation scheme actually create?”

What each scheme is testing

Different cross-validation schemes answer different questions, and this matters for how results should be interpreted. Leave-one-timepoint-out tests how well the model predicts a single omitted time point when all others are available, a relatively gentle task when neighboring time points are close together. Leave-contiguous-block-out tests whether the model can bridge an entire missing time window, a harder task that is more sensitive to gaps in the profile. Ordered K-fold sets aside larger ordered portions of the dataset at a time. Shuffle split sets aside randomly selected time points without regard to their order, which can serve as a general check of overall curve shape but may make the model appear more reliable than it is when nearby time points carry similar information.

These distinctions matter because adjacent time points in a degradation profile are often similar to one another, measurement errors may be related across time, and early, middle, and late portions of a degradation profile can reflect qualitatively different behavior. Standard cross-validation procedures are reviewed in [13]; additional considerations for temporally ordered or dependent data are discussed in [14, 15, 16].

Interpreting results for small IVIVC datasets

For the small datasets typical in IVIVC development, cross-validation results should be treated as an exploratory diagnostic or sensitivity analysis rather than a definitive performance benchmark. Compare the CV metric with the corresponding metric for the model fitted to the full dataset, but also consider whether the chosen splitting pattern reflects the practical prediction question of interest.

For example, strong leave-one-timepoint-out performance alongside weaker leave-contiguous-block-out performance may indicate that the model can interpolate individual missing points well, but becomes less reliable when an entire local region of the profile is absent from fitting. That distinction may or may not matter depending on the intended use of the model.

Fit/CV ratios

The app reports fit/CV ratios to summarize the relationship between full-fit and cross-validation performance for the same metric. A ratio near 1 indicates that the two are similar. A ratio far from 1 may suggest reduced generalizability, overfitting, model instability, or sensitivity to the splitting scheme. Because the direction of concern depends on the metric (for example, a ratio above 1 is concerning for RMSE but not for R^2), the ratio should always be interpreted alongside the raw fit and CV values, the chosen splitting scheme, and the fitted plots.

Cross-validation comparison badges

The app displays cross-validation comparison badges as interpretive aids. Badges are assigned based on the preferred direction of the metric: for lower-is-better metrics such as RMSE, a worse CV result means the CV value is larger than the full-fit value; for higher-is-better metrics such as R^2 , a worse CV result means the CV value is smaller. The thresholds used are:

- **Similar:** CV performance is within a factor of 1.25 of the full-fit value.
- **Moderately worse:** CV performance is 1.25- to 2-fold worse than the full-fit value.
- **Substantially worse:** CV performance is more than 2-fold worse than the full-fit value.

These thresholds are heuristic review aids to help flag results that warrant closer attention. They are not statistical acceptance criteria or regulatory standards.

Metrics excluded from cross-validation comparisons

The app does not include adjusted R^2 , information criteria (AIC, AIC_c, and BIC), or normalized RMSE (NRMSE) in cross-validation comparison tables. The reason is that these metrics are not directly comparable across fits that use different numbers of observations:

- **Adjusted R^2 and information criteria** incorporate the number of observations or fitted parameters as part of their calculation. Because the final model and each cross-validation model are fitted to different subsets of the data, these quantities are not on the same scale and should not be compared directly.
- **NRMSE** is normalized by a factor that generally differs across subsets. This means the final-fit and CV values may not share the same denominator, making a direct comparison misleading.

Use the eligible cross-validation metrics to assess prediction behavior during CV. The ineligible CV metrics can still be used to evaluate and compare fitted models. Note that not all eligible metrics are available for every cross-validation scheme. Some metrics require more than one observation to be defined or stable. For example, R^2 cannot be calculated for a single omitted point and will be unavailable under leave-one-timepoint-out CV. Where a metric is unavailable for the selected scheme, rely on the metrics that are reported.

A.6 Interpreting residual diagnostics

Residuals are the differences between observed and predicted values (observed minus predicted). Residual diagnostics help assess whether model errors show systematic patterns, changes in error size over time (heteroscedasticity), or other departures from model assumptions [23]. The app provides two qualitative diagnostic plots: residual-vs-time plots and quantile-quantile (Q-Q) plots.

Residual-vs-time plots

Residuals are plotted in the same units as the fitted response. To make visual comparison across models easier, all comparable models within an approach share a common y-axis scale, derived from the pooled 95th percentile of absolute residuals across those models. Two reference bands are drawn on each plot:

- A **shaded band** at $\pm 25\%$ of the common residual scale.
- **Dashed lines** at $\pm 50\%$ of the common residual scale.

These bands provide visual context for comparing residual magnitude across models. They are not confidence intervals, prediction intervals, acceptance limits, or regulatory criteria. Because the common scale is derived from the selected models, it will shift if the model set changes.

Q-Q plots

Q-Q plots compare the ordered standardized residuals against the pattern expected under a normal distribution. Points near the diagonal reference line are broadly consistent with normal residual variation. Systematic curvature, large departures in the tails, or isolated extreme points may suggest non-normality, outliers, or a model form that does not fit the data well. Because IVIVC datasets typically contain few time points, Q-Q plots should be treated as exploratory diagnostics rather than pass/fail tests.

What to look for

Residual diagnostics should be reviewed qualitatively—by eye—rather than by formal statistical tests. Although statistical tests exist for common residual problems such as non-normal errors, correlated errors across time (auto-correlation), and variance that changes across the response range (heteroscedasticity), these tests require more data than IVIVC datasets typically provide. With few observations, such tests often fail to detect real problems or flag spurious ones, making visual review more reliable in practice. Residuals scattered around zero in the residual-vs-time plot, and points near the diagonal in the Q-Q plot, are consistent with an appropriate model form, though they do not prove the model is valid. The following patterns may indicate bias or a model form that does not fully capture the data:

- Sustained curvature or systematic trends in the residuals over time.
- Clustering, where residuals at adjacent time points are consistently on the same side of zero.
- Repeated excursions outside the reference bands.

When any of these patterns appear, consider whether a different model form, additional parameters, or a transformed response might better describe the underlying relationship.

A.7 Interpreting parameter uncertainty

It is important to check whether the available data can estimate model parameters reliably. The app reports each parameter estimate $\pm 95\%$ confidence interval. The plus-minus value represents the confidence-interval half-width around the estimate. In practical terms, a wide interval means the data do not pin down that parameter precisely, while a narrow interval means the estimate is more stable.

The interval is calculated from the nonlinear least-squares covariance matrix and residual degrees of freedom. It summarizes parameter uncertainty implied by the fitted model and should be interpreted cautiously, especially for nonlinear models and small datasets.

The app summarizes parameter uncertainty using informational badges as screening aids, not acceptance criteria. They flag fits that may deserve closer review before the model is used for interpretation or prediction.

Diagnostic	What it summarizes	Interpretation guidance
Relative standard error	The uncertainty in a parameter estimate expressed as a percentage of the estimate itself.	A large value means the uncertainty is large relative to the estimate, so the data are not pinning down that parameter well. The app flags values of at least 50% as elevated and at least 100% as large.
Parameter coupling / correlation	How strongly pairs of parameter estimates move together during fitting (pairwise correlation).	When two parameters are highly correlated, the model can reach a similar fit by trading off one against the other, making both harder to estimate reliably. Some coupling is normal in nonlinear models. The app reports correlation alone as <i>parameter coupling noted</i> ; when high correlation coincides with elevated relative standard error or a numerical sensitivity flag, it reports a stronger identifiability caution. Absolute correlations of at least 0.90 are noted, with stronger concern when correlations of at least 0.95 occur alongside other uncertainty diagnostics.

If no parameter uncertainty concerns are flagged, the app did not detect large relative standard errors or strong parameter coupling under these screening checks. This does not validate the model, confirm mechanistic correctness, or establish regulatory acceptability. Conversely, a coupling or warning badge does not automatically invalidate a model; rather, it is a prompt to review the flagged parameters alongside the fitted curves, residuals, cross-validation behavior, and scientific plausibility.

Parameter uncertainty deserves particular attention when a model will be interpreted mechanistically, used to extrapolate beyond the observed data range, or used to compare degradation behavior across materials, processes, or test conditions. Strong fit metrics do not compensate for parameters that are poorly estimated or scientifically implausible.

A.8 Considering model complexity

More complex models can fit observed data more closely, but with small datasets this can be a liability rather than an asset. A flexible model may trace the noise in the data rather than the underlying pattern, and its parameters may be poorly estimated as a result, an issue called *overfitting*. Adjusted R^2 , AIC, AIC_c, BIC, cross-validation metrics, and residual diagnostics can all help assess whether added complexity is justified [24].

When two models have similar fit and similar empirical support, the simpler one is usually easier to justify, if it is scientifically plausible and adequate for the intended use.

A.9 Scientific plausibility and intended use

Statistical performance alone is not sufficient to evaluate a model. A model should also make sense given what is known about the polymer, degradation mechanism, and test conditions. A model that is statistically competitive but produces a fitted shape, parameter values, or extrapolated behavior that conflicts with known degradation behavior may be difficult to justify regardless of its metrics.

The intended use should also guide how much supporting evidence is needed. Exploratory model development can tolerate more uncertainty than a model intended to support a regulatory argument, a design change, or a prediction of in vivo performance.

A.10 Suggested model review workflow

1. Confirm that the dataset, preprocessing choices, and selected response values (the measured values being modeled) are appropriate for the intended analysis.
2. Review fitted curves and residual plots for each candidate model, checking for systematic patterns or implausible fitted shapes.
3. Compare goodness-of-fit metrics such as R^2 and RMSE in the context of assay variability, endpoint importance, and intended use.
4. Review cross-validation performance and fit/CV ratios to assess sensitivity to the available data.
5. Review relative model evidence using information criteria to compare fit quality against model complexity.
6. Review fitted-parameter estimates, 95% confidence intervals, uncertainty diagnostics, and scientific plausibility.
7. Consider whether model complexity is justified by the available data and the degradation mechanism.
8. Document the user-selected final model based on a balance of statistical performance, scientific plausibility, and intended use.

A.11 Questions to guide model review

The following questions are intended as practical prompts for reviewing model results, not as formal acceptance criteria.

A.11.1 Does the model describe the observed data well?

- Does the fitted curve follow the main trend in the data?
- Are performance metrics such as R^2 , RMSE, and NRMSE reasonable for the intended use?
- Is the prediction error small enough to be meaningful relative to assay variability, measurement uncertainty, or the degradation endpoint being evaluated?
- Are there regions where the model consistently overpredicts or underpredicts?

A.11.2 Does the model generalize beyond the fitted data?

- Are cross-validation metrics similar to the metrics for the model fitted to the full dataset, and are fit/CV ratios near 1?
- Could the cross-validation result be sensitive to the splitting scheme, number of observations, or a small number of influential points?
- If cross-validation performance is poor, is the model still useful as an exploratory description, or would a simpler or more stable model be more appropriate?

A.11.3 Are the model parameters interpretable and stable?

- Are the fitted parameter values scientifically plausible?
- Are the 95% confidence intervals narrow enough that the parameter estimates appear reasonably precise?
- Are any parameters strongly coupled—meaning different combinations of parameter values could produce a similar fitted curve?
- If coupling is present, does it coincide with large parameter uncertainty or an identifiability caution?
- Are individual parameter values being interpreted mechanistically, or is the model being used mainly to describe the overall curve shape?

A model can have a visually reasonable fitted curve while still having uncertain or coupled parameters. In that case, the curve may be useful for empirical description within the observed data range, but individual parameter values should be interpreted cautiously.

A.11.4 Is the model unnecessarily complex?

- Does the more complex model improve fit, cross-validation behavior, or relative model evidence enough to justify the added parameters?
- Do AIC, AIC_c, or BIC support the additional complexity within the candidate model set?
- If two models perform similarly, is the simpler one easier to explain and justify?
- Does the dataset contain enough observations to support the number of fitted parameters?

A.11.5 Is the model scientifically plausible?

- Is the fitted curve shape consistent with expected polymer degradation behavior?
- Are the model assumptions reasonable for the material, test conditions, and degradation endpoint?
- Does the model behave plausibly near the edges of the observed data range, and would it remain plausible if used for interpolation or extrapolation?

A statistically competitive model may still be difficult to justify if its shape, parameter values, or extrapolated behavior conflict with the known degradation process.

A.11.6 Is the model appropriate for the intended use?

- Is the model being used for exploratory analysis, interpolation, prediction, comparison across conditions, or support for a regulatory rationale?
- Is the level of uncertainty acceptable for that purpose?
- Are the model's limitations clearly documented?
- Would an independent reviewer understand why this model was selected over reasonable alternatives?

A final model rationale should address fitted performance, cross-validation behavior, relative empirical support, parameter diagnostics, residual behavior, scientific plausibility, and intended use.

A.12 Example interpretation statements

The following examples illustrate wording that may be adapted for reports. Users should tailor the language to the specific analysis and supporting evidence.

- Model A had the lowest RMSE and highest adjusted R^2 among the evaluated models, indicating the strongest goodness-of-fit under the selected preprocessing conditions.
- Model A was selected over the more complex Model B because both models showed similar fit and AIC_c support (Delta AIC_c < 2), indicating that the additional complexity of Model B was not justified by the data.
- Although Model C had strong goodness-of-fit, its cross-validation error was substantially higher than its fitted error, suggesting sensitivity to which observations were included in fitting.
- Residuals for Model A were scattered around zero without obvious systematic patterns, consistent with an adequate model form for the observed data.
- Model B showed a systematic trend in the residuals across time, suggesting the selected model form may not fully capture the degradation profile.
- The fitted parameters for Model A were within plausible ranges for the degradation mechanism, with narrow confidence intervals indicating stable estimation.
- Model C showed strong parameter coupling between the rate and shape parameters, with relative standard errors exceeding 100%. Parameter estimates were retained for curve description but were not interpreted individually.
- The selected model was chosen based on a balance of goodness-of-fit, cross-validation behavior, relative empirical support, parameter stability, and scientific plausibility for the degradation mechanism and intended use.

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